

CAMPUS INCUBATION & STUDENT INNOVATION CELL · KL UNIVERSITY

Bittu Xerox Kiosk

24/7 Autonomous Smart Document Station for University Hostels & Campus Blocks

₹48,420

TOTAL CAPEX BOM

75% cheaper than commercial vendor kiosks

0.36 m²

MICRO FOOTPRINT

Fits directly in hostel and academic corridors

100% UPI

ZERO-CLERK FLOW

Dynamic QR & OTP PIN instant checkout

100% RAM

ZERO-CLERK PRIVACY

Ephemeral spooling with auto Linux memory wipe

Founder & Hardware Lead: Sai Nikhil Vukka | Dept. of CSE, KL University

Project Repository: github.com/Sai0Nikhil/Xerox_kisok

The Midnight Printing Bottleneck Across Indian University Campuses

15,000+ students face daily friction, curfew lockouts, and critical data privacy vulnerabilities.

5:30 PM CURFEW LOCKOUT

Late-Night Document Deserts

- Campus stationery shops strictly close by 5:30 PM.
- Hostel students working on midnight assignments, lab records, and exam prep have zero printing access.
- Forces risky off-campus travel or last-minute morning panics.

Impact: 0 Available Options after 8 PM

30-45 MIN MORNING QUEUES

Peak Submission Bottlenecks

- 8:00 AM - 9:00 AM submission rush overwhelms manual operators.
- Students miss morning lectures waiting for 2-page printouts.
- Manual UPI verification and cash change slows every transaction.

Impact: 40+ Mins Average Peak Wait

ZERO PRIVACY ON SHARED PCS

Aadhaar & Resume Data Leakage

- Documents sent via WhatsApp Web or opened on shared shop desktops.
- Aadhaar, PAN, resumes, and medical certificates remain in 'Downloads' folders indefinitely.
- Complete violation of student digital privacy & DPDP guidelines.

Impact: 100% Unencrypted Shop Exposure

Enterprise-Grade Document Security vs. Traditional Xerox Shops

Guaranteed end-to-end data privacy with ephemeral volatile RAM processing.

TRADITIONAL CAMPUS XEROX SHOP

High Privacy Risk & Manual Exposure

- ✗ Public WhatsApp Web login exposes personal chat previews and contacts.
- ✗ Files downloaded to persistent HDD 'Downloads' folder and never purged.
- ✗ Next student in queue or operator can easily browse sensitive personal files.
- ✗ Identity theft risk: Government IDs, grade sheets, and bank forms left exposed.
- ✗ Human operator bottleneck: 2–3 minutes spent per student just finding files.

Vulnerability: High Exposure & Persistent Storage

BITTU ZERO-CLERK KIOSK PIPELINE

Direct Volatile RAM & Instant Shredding

- ✓ End-to-End TLS 1.3 encrypted direct upload from student's phone browser.
- ✓ Stored exclusively in Linux tmpfs (RAM disk) — NEVER written to physical SD/SSD storage.
- ✓ Rendered directly to CUPS laser engine; post-print Linux shred wipes memory blocks instantly.
- ✓ Zero human intermediary: Operator never sees, opens, or stores student documents.
- ✓ Compliant with Indian Digital Personal Data Protection (DPDP) standards.

Guarantee: 0 Bytes Persistent Storage | ISO 27001 Aligned

Robust, Modular Hardware Station Built on 0.36 m²

Commercial-grade off-the-shelf (COTS) components engineered for 24/7 heavy-duty campus operation.

COMPUTE CORE

Raspberry Pi 4B + 15.6" Touch

- Dedicated Linux CUPS print server & telemetry agent.
- 15.6" 1080p capacitive responsive touch display.
- Multi-language UI (English, Telugu, Hindi).

Budget: ₹14,500

PRINT ENGINE

Brother DCP-L2541DW Laser

- 30 PPM high-speed printing with auto-duplexing.
- 2,600-page high-capacity toner cartridge (TN-2365).
- 250-sheet enclosed anti-dust paper tray.

Budget: ₹18,500

SCANNING BAY

Canon CanoScan LiDE 300

- 2400 x 2400 DPI optical scanning resolution.
- Fast 10-sec A4 color scanning into PDF/JPEG.
- Secure recessed drawer with mechanical lock.

Budget: ₹4,200

POWER & CHASSIS

600VA UPS + Steel Enclosure

- 30-minute power backup prevents print-job aborts.
- Dual 120mm PWM fans prevent thermal throttling.
- Modbus RS485 energy meter + keyed access.

Budget: ₹11,220

Frictionless 4–Step Self–Service Flow in Under 45 Seconds

Intuitive mobile web app interface — No mandatory mobile app download required.

STEP 01

Scan Dynamic QR

Student scans the dynamic screen QR code using any smartphone camera or UPI app, instantly opening the mobile web uploader without app download.

Speed: < 5 Seconds

STEP 02

Upload & Configure

Uploads PDF, Word, or image directly via HTTPS. Selects color/B&W, single or auto–duplex, and page ranges with real–time interactive preview.

Speed: < 15 Seconds

STEP 03

1–Tap UPI Pay

Calculates exact fare. Student pays via GPay, PhonePe, or Paytm. Webhook verifies payment in 500ms and immediately unlocks the print job.

Speed: < 10 Seconds

STEP 04

Instant Print & Shred

Laser engine prints document into collection bin. Linux spooler immediately zeroes volatile RAM buffer. Student grabs fresh printout.

Speed: < 12 Seconds

Industrial IoT Telemetry for 99.8% Unattended Uptime

Automated consumable tracking, predictive jam prevention, and remote energy auditing.

OPTICAL PAPER & TONER SENSING

Zero Stock-Out Guarantee

- Ultrasonic & optical tray sensors measure paper depth in real-time.
- Software page counters calculate precise toner life (TN-2365 cartridge).
- Automated WhatsApp / SMS dispatch alerts campus hostel warden when stock hits 15%.
- Ensures kiosk never accepts payment when out of consumables.

99.8% Uptime SLA

THERMAL & JAM PROTECTION

Smart Health & Diagnostics

- Internal thermistors control dual 120mm exhaust fans dynamically to prevent heat buildup.
- CUPS error interceptor halts queue on paper misfeed and auto-refunds UPI in 3 seconds.
- Line-Interactive 600VA UPS buffers voltage spikes and allows clean job completion.

Auto 3-Sec UPI Refund

CENTRALIZED MULTI-KIOSK FLEET

Remote Cloud Monitoring

- Lightweight MQTT / WebSocket daemon sends heartbeat telemetry every 30 seconds.
- Central dashboard tracks daily prints, paper consumption, and revenue per hostel block.
- Secure SSH & OTA firmware updates deployed remotely without opening the chassis.

Real-time Fleet Control

Realistic Cost Structure (Strict, Non-Overestimated Pricing)

Sustainable unit economics based on verified Indian wholesale consumable prices.

COST & REVENUE BREAKDOWN

Per-Page Economics (A4 B&W Single-Sided)

75 GSM Premium A4 Paper (JK / Century ream ₹260/500)	₹0.52
High-Yield Compatible Toner (ProDot TN-2365 ₹650 / 2,600 pp)	₹0.25
Laser Drum Unit Wear & Maintenance (12k lifespan)	₹0.10
Campus Electricity & UPS Loss (180W avg @ ₹8/unit)	₹0.03
Hostel/Campus Revenue Share & Reserve Fund	₹0.25
TOTAL DIRECT OPERATING COST	₹1.15 - ₹1.20
STUDENT SELLING PRICE (Market standard)	₹2.50 / page
NET OPERATING PROFIT PER PAGE	₹1.30 (52%)

52% Net Profit Margin

FINANCIAL SUSTAINABILITY

Key Financial Pillars

- **52% Healthy Net Margin:** Generates ₹1.30 net profit per single-sided print, leaving ample buffer for machine wear and paper price fluctuations.
- **Auto-Duplex Incentive:** Students get 2-sided print for ₹3.50 (saving ₹1.50). Conserves 40% paper stock while yielding ₹1.68 net profit.
- **Zero Bad Debt / Credit:** 100% upfront UPI collection guarantees zero unpaid printing tabs or coin handling losses.
- **Low Hardware CAPEX:** Total kiosk hardware budget ₹48,420 enables rapid break-even within 4 to 10 months.

Total BOM CAPEX: ₹48,420

Strict Capital Recovery Across 3 Campus Deployment Scenarios

CAPEX Recovery Model against ₹48,420 Hardware Investment across different density tiers.

CONSERVATIVE TIER

Govt College Corridor

- **Daily Volume:** 125 pages/day
- **Monthly Volume:** 3,750 pages
- **Gross Revenue:** ₹9,375 / month
- **Monthly Op Cost:** ₹4,500
- **Monthly Net Profit:** ₹4,875 / month

Capital Payback: 10.5 Months

REALISTIC TARGET

Private Hostel (1,200 beds)

- **Daily Volume:** 300 pages/day
- **Monthly Volume:** 9,000 pages
- **Gross Revenue:** ₹22,500 / month
- **Monthly Op Cost:** ₹10,800
- **Monthly Net Profit:** ₹11,700 / month

Capital Payback: 4.2 Months

PEAK DENSITY TIER

Academic Central Block

- **Daily Volume:** 600 pages/day
- **Monthly Volume:** 18,000 pages
- **Gross Revenue:** ₹45,000 / month
- **Monthly Op Cost:** ₹21,600
- **Monthly Net Profit:** ₹23,400 / month

Capital Payback: 2.1 Months

Why Bittu Kiosk Wins Over Commercial Printing Franchises

Unmatched unit economics, open hardware architecture, and zero franchise royalty fees.

COMMERCIAL FRANCHISES

High CAPEX & Vendor Lock-In

- ✗ Hardware Cost: ₹1.65 Lakh – ₹3.50 Lakh
- ✗ Franchise Royalty: 20% – 30% cut on every print
- ✗ Locked proprietary cartridges at 3x market rate
- ✗ Slow response time for hostel repairs
- ✗ Rigid closed software with recurring fees

Verdict: High Financial Risk

MANUAL CAMPUS STALLS

Curfew & Privacy Flaws

- ✗ Hours: Strictly 9:00 AM – 5:30 PM (0 midnight access)
- ✗ Privacy: Files exposed on public desktop & WhatsApp Web
- ✗ Queue Time: 30 – 45 minute morning submission queues
- ✗ High manpower overhead for paper handling
- ✗ Manual cash/change bottlenecks & lost tabs

Verdict: Poor Student Experience

BITTU XEROX KIOSK

Lean, Open & 24/7 Scalable

- ✓ Hardware Cost: ₹48,420 (75% lower initial CAPEX)
- ✓ Franchise Royalty: 0% — 100% retained by campus
- ✓ Open COTS Brother/Canon standard supplies
- ✓ Availability: 24/7/365 continuous unattended operation
- ✓ Privacy: Ephemeral RAM spooler + Auto-Shred compliance

Verdict: Optimal Campus Fit

Seamless Integration with University ERP & WhatsApp Ecosystem

Purpose-built digital workflows tailored for Indian collegiate academic operations.

1-TAP EXAM HALL TICKETS

Direct University ERP Bridge

- Integrates with University Student Portals (e.g., KL ERP).
- Authenticated students can print watermarked mid-term & semester hall tickets with 1 touch.
- Eliminates admin office queues before semester exams.

Advantage: Zero Pre-Exam Queue

WHATSAPP DOCUMENT BOT

Remote OTP PIN Express

- Students forward document from hostel bed to Bittu WhatsApp Bot.
- Bot replies with dynamic 4-digit PIN code valid for 2 hours.
- Student walks up, punches PIN on touch screen, and collects document in 10 seconds.

Advantage: Instant Walk-Up Collect

MULTILINGUAL ACCESSIBLE UI

Inclusive Student Interface

- Intuitive GUI designed with large, high-contrast touch targets.
- Full multilingual language switcher supporting English, Telugu, and Hindi.
- Audio cue confirmation upon successful print tray delivery.

Advantage: 100% Inclusive Reach

60-Day Pilot Deployment Plan & University Incubation Support

Clear roadmap from workshop prototype to live hostel testbed.

60-DAY PILOT MILESTONES

Execution Timeline

- **Days 01 – 20 (Procurement & Fabrication):** Procure COTS hardware (Brother printer, Pi 4, UPS) and fabricate 1.2mm CRCA steel chassis via local laser cutting & powder coating.
- **Days 21 – 40 (Firmware & Payment Integration):** Install headless Linux CUPS, ephemeral RAM spooling daemon, and integrate PhonePe/UPI dynamic QR webhook gateway.
- **Days 41 – 60 (Live Pilot in KL Boys Hostel):** Deploy Kiosk Unit 01 in KL University Boys Hostel ground corridor serving 1,200 students for real-world load & stress testing.

Target: Live Hostel Rollout by Day 60

INCUBATION & CAMPUS SUPPORT

The Support Requested

- **0.36 m² Physical Space Permission:** A small 4 sq.ft footprint near the hostel warden office / entrance lobby.
- **Standard Electrical & Wi-Fi Point:** 1 single 5A/15A wall power socket + campus Wi-Fi / LAN connection.
- **₹48,420 Prototype Seed Grant:** Direct hardware procurement grant for Unit 01 components (BOM verified in procurement sheet).
- **Pilot Mentorship & Campus Pilot Pass:** Support from KL Incubation Center for legal student testing & campus feedback.

Total Seed Grant Ask: ₹48,420

SMART CAMPUS INFRASTRUCTURE · STUDENT INNOVATION

Autonomous. Private. 24/7.

Solving the campus printing crisis with modern, privacy-first, zero-clerk technology.

PROJECT INITIATIVE & CONTACT DETAILS

Project Lead: Sai Nikhil Vukka

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Project GitHub: github.com/SaiONikhil/Xerox_kisok

Engineering Stack: Raspberry Pi 4B · Brother Auto-Duplex · CRCA Steel

Pilot Goal: Unit 01 Deployment in KL University Hostels